I Model Details

I.A Merchant Profits and Optimal Pricing

Merchant profits equal the integral of per-unit margins times quantities across consumer types:

$$\Pi\left(\gamma, p, M\right) = \int \sum_{w \in \mathcal{W}} \tilde{\mu}_y^w \times q^w\left(\gamma, p, M, y\right) \times L_M^w\left(\gamma, p\right) \, \mathrm{d}F\left(y\right) \tag{1}$$

where $\tilde{\mu}_y^w$ is the mass of consumers at income level y with wallet w.

Per-unit margins. The average margin L_M^w weights each instrument by its share of sales:

$$L_M^w\left(p\right) = \underbrace{\frac{1 - \pi_{M,w_1}^w - \pi_{M,w_2}^w}{1 + \gamma v_M^w}}_{\text{share on cash}} \times \left(p - 1\right) + \sum_{j=1}^{2} \underbrace{\frac{\pi_{M,w_j}^w\left(1+\gamma\right)}{1 + \gamma v_M^w}}_{\text{share on card } w_j} \times \left(p\left(1 - \tau_{w_j}\right) - 1\right) \tag{2}$$

Collecting terms yields

$$L_M^w\left(p\right) = p\left(1 - \frac{(1+\gamma)\tau_M^w}{1 + \gamma v_M^w}\right) - 1 \tag{3}$$

where $\tau_M^w = \sum_{j=1}^{J} \pi_{M,j}^w \tau_j$ is the wallet-average fee.

Profits. CES preferences are homothetic, so $q^w(\gamma, p, M, y) = q^w(\gamma, p, M, 1) \cdot y$; the income distribution then integrates out of Π. Using $\tilde{\mu}^w q^w(\gamma, p, M, 1) = C\mu^w p^{-\sigma}\left(1 + \gamma v_M^w\right)$ (with C the aggregate demand constant and μ^w the demand shares from Equation **??**). Merchant profits become:

$$\Pi\left(\gamma, p, M\right) = C \times \sum_{w \in \mathcal{W}} \mu^w\left(1 + \gamma v_M^w\right) p^{-\sigma} \times \left[p\left(1 - \frac{(1+\gamma)\tau_M^w}{1 + \gamma v_M^w}\right) - 1\right] \tag{4}$$

This matches Equation **??** in the main text.

Optimal price. Setting $\partial \Pi / \partial p = 0$:

$$\sum_{w \in \mathcal{W}} \mu^w\left(1 + \gamma v_M^w\right)\left[\left(1 - \sigma\right) p\left(1 - \frac{(1+\gamma)\tau_M^w}{1 + \gamma v_M^w}\right) + \sigma\right] = 0 \tag{5}$$

Solving for p gives the optimal price in Equation **??**.

I.B Linearizing Merchant Profits

Define quasi-profits $\overline{\Pi}$ by:

$$\overline{\Pi}\left(\gamma, M, \tau\right) \equiv C \times \left(\frac{\sigma}{\sigma-1}\right)^{1-\sigma} \left\{-a_M + b_M\gamma + \frac{1}{\sigma}\right\} \tag{6}$$

where C is the aggregate demand constant from Equation **??**, and

$$a_M = \sum_{w\in\mathcal{W}} \mu^w \tau_M^w, \qquad b_M = \frac{1}{\sigma}\sum_{w\in\mathcal{W}} \mu^w \left(v_M^w - \sigma\tau_M^w\right) \tag{7}$$

recovering the coefficients in Equation **??**. The v_M^w component captures incremental volume; the $-\sigma\tau_M^w$ captures lost margin due to fees.

Theorem 1. *For any γ, M, and τ with $\tau^{\max} \equiv \max_j \tau_j$,*

$$\widehat{\Pi}\left(\gamma, M, \tau\right) - \overline{\Pi}\left(\gamma, M, \tau\right) = \left(1+\gamma\right) O\left(\left(\tau^{\max}\right)^2\right)$$

Proof. The optimal payment-specific prices are $p_j = \frac{\sigma}{\sigma-1}\frac{1}{1-\tau_j}$ for each payment method j. By the envelope theorem, any price within $O(\tau_j)$ of p_j delivers the same profit up to second-order terms in τ_j. Setting $\overline{p} = \frac{\sigma}{\sigma-1}$ and collecting terms recovers the linear approximation. Collecting terms in γ:

$$\Pi\left(\gamma, \overline{p}, M\right) = C \times \left(\frac{\sigma}{\sigma-1}\right)^{1-\sigma} \left(-a_M + b_M\gamma + \frac{1}{\sigma}\right) = \overline{\Pi}\left(\gamma, M, \tau\right)$$

□

Online Appendix **??** confirms that $\overline{\Pi}$ is numerically accurate.